



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-2024
HALF YEARLY EXAMINATION

CLASS:X
SUBJECT: ENVIRONMENTAL SCIENCE

FULL MARKS: 80
TIME:2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.*

*Attempt all questions from Section A and any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of five printed pages.*

SECTION A (40 Marks)
Attempt all questions from this Section

QUESTION 1

[16]

Choose the correct answers to the questions from the given options.
(Do not copy questions, write the correct answers only.)

- i. Religious beliefs, preference of a male child, lack of family planning are factors responsible for increase in population in which countries?
 - a) Developed countries
 - b) Developing countries
 - c) All countries
 - d) None of the countries
- ii. Land – use planning purpose is to _____
 - a) Use of land and reduce its quality
 - b) Avoid misuse of land and enhance its quality
 - c) Use of land and enhance its degradation
 - d) Enhance land degradation and misuse of land
- iii. Irrigation that involves adding water to the plant drop by drop at or near the roots is known as
 - a) Trickle drip irrigation
 - b) Macro irrigation
 - c) Conservation tillage
 - d) Regenerative technique
- iv. Eucalyptus restricts the growth of other plants nearby because of
 - a) sub soiling
 - b) agrostological effect

- c) allelopathic effect
 - d) meteorological effect.
- v. Which of the following is an advantage of zoning strategy?
 - a) Can increase the cost of building new structures in some areas.
 - b) The government and private enterprises can lose control over its land use.
 - c) Can discourage some development in some areas.
 - d) Can conserve environmentally sensitive areas.
- vi. Techniques which focus strongly on rebuilding of soils and restoring ecosystems is
 - a) Intercropping
 - b) Crop rotation
 - c) Agroforestry
 - d) Regenerative farming
- vii. PNG stands for
 - a) Portable Natural Gas
 - b) Piped Natural Gas
 - c) Piped Nitrogen Gas
 - d) Piped Nature Gas
- viii. Causes of excessive vehicular pollution include
 - a) New vehicles, Lack of effective Rapid public transport system, Improper traffic management
 - b) Inadequate inspection of vehicles, Improper traffic management, Lack of effective Rapid public transport system
 - c) Good Road conditions, Lack of effective Rapid public transport system, Improper traffic management
 - d) Improper traffic management, good roads, inspection of new vehicles
- ix. With reference to the figure, answer the following question



- The figure given above is a Convention to save
- a) Wetland
 - b) River
 - c) Ocean
 - d) Desert
- x. Assertion(A)--Three quarters of the world's population is inadequately fed, so there is hunger and malnutrition in the developing countries.
 Reason (R)—The factors of food shortage in the developing countries are poverty, unsuccessful agriculture and unequal distribution in food supply.
 Which of the following is correct?

- a) A is true and R is not the correct reason for A.
 b) A is false and R is not the correct reason for A.
 c) A is false but R is true.
 d) Both A and R are true.
- xi. **Statement A:** GM crops have reduced food insecurity especially in the developing countries.
Statement B: GM crops need pesticides which lead to emergence of super pests and super weeds.
- a) Both Statements A and B are correct.
 b) Statement A is correct and Statement B is incorrect.
 c) Statement A is incorrect and Statement B is correct.
 d) Both statements A and B are incorrect.
- xii. Match Column 1 with Column 2 and choose the correct option from those given below.

Column 1

- (A) Carbon dioxide
 (B) Carbon monoxide
 (C) Photochemical Smog
 (D) Marble Cancer
 (E) Cholera

Column 2

- (i) Ozone
 (ii) Global warming
 (iii) Carboxyhaemoglobin
 (iv) Contaminated water
 (v) Acid rain

- a) (A-iii), (B-v), (C-i), (D-ii), (E-iv)
 b) (A-ii), (B-iii), (C-i), (D-v), (E-iv)
 c) (A-iv), (B-ii), (C-iii), (D-v), (E-i)
 d) (A-i), (B-iii), (C-v), (D-ii), (E-iv)

The following questions (xiii—xvi) are based on the figure given below. Answer these questions by choosing the correct option.



- xiii. The zone shown in the picture is approximately
- a) 10,200 Km
 b) 12,700 Km

- c) 10,400 Km
 - d) 12,400 Km
- xiv. Three heritage sites included in TTZ.
- a) Taj Mahal, Agra Fort and Fatehpur Sikri
 - b) Taj Mahal, Fatehpur Sikri and India Gate
 - c) Taj Mahal, Agra fort and Red Fort
 - d) Taj Mahal, Fatehpur Sikri and Red fort
- xv. Due to which of the following pollutants is the Taj Mahal threatened?
- a) Carbon dioxide
 - b) Carbon monoxide
 - c) Sulphur dioxide
 - d) Ozone
- xvi. Clean fuels that can be used in the said area are
- a) CNG, Petrol, Kerosene, Ethanol
 - b) CNG, Propane, Ethanol, Oxygenated fuels
 - c) CNG, Diesel, Ether, Petrol
 - d) CNG, Diesel, Kerosene, Petrol

Question 2

- a) How do forests maintain climate stability? [2]
- b) State one advantage and one disadvantage of captive breeding. [2]
- c) Which gas is the main constituent of Biogas? Give one advantage of using Biogas. [2]
- d) What are bunds? How do they reduce soil erosion? [2]

Question 3

- a) List two advantages of in-situ conservation. [2]
- b) Mention any two ways by which industrial development can be controlled. [2]
- c) Discuss any two ways of combating deforestation. [2]
- d) Mention any two benefits of Green Manure. [2]

Question 4

- a) State any two causes for loss of wildlife habitat. [2]
- b) What are energy plantations? How are they beneficial? [2]
- c) Give the full form of CITES and IUCN. [2]
- d) State the difference between core and buffer zone of a Biosphere Reserve. [2]

SECTION B (40 Marks) **Attempt any four questions from this Section**

Question 5

- a) 'Sukhomajri is a successful model of community participation for ecological restoration'.
Give a brief background of the problems faced by the people of Sukhomajri

- and the steps taken by the community for ecological restoration. [5]
- b) Explain the farming techniques used to reduce soil erosion. [5]

Question 6

- a) 'IPM is a sustainable approach to managing pests'. Comment on the various principles of managing pests in IPM approach. [5]
- b) Write a note on women centered human development as a strategy for controlling the population growth. [5]

Question 7

- a) "Overgrazing is a menace". Give three impacts of overgrazing and two measures to control the menace. [5]
- b) "In most cities increasingly CNG is used as a fuel in vehicles". Give appropriate reasons for your answer. [5]

Question 8

- a) What is Conservation Tillage? Mention three advantages of Conservation Tillage. [5]
- b) "Growth of slums is a common feature of Urbanisation." State five disadvantages of the growth of slums. [5]

Question 9

- a) What is the ecological impact of population growth? [5]
- b) What do you understand by ex-situ conservation of biodiversity? Give two advantages and two disadvantages of ex-situ conservation? [5]

Question 10

Land is the most important component of agriculture. 60% of our vast population of over one billion depends on agriculture for survival. Agricultural productivity depends, to a great extent, on the rights a cultivator has on the land and its produce. Independent India took bold steps in land reform. However, inspite of the benefits of land reforms, it has not been uniformly successful throughout the country.

- a) Which movement was associated with land reform? Mention the basic features of land reform. [5]
- b) What were the reasons behind Land Reform not being uniformly successful throughout the country? [5]